

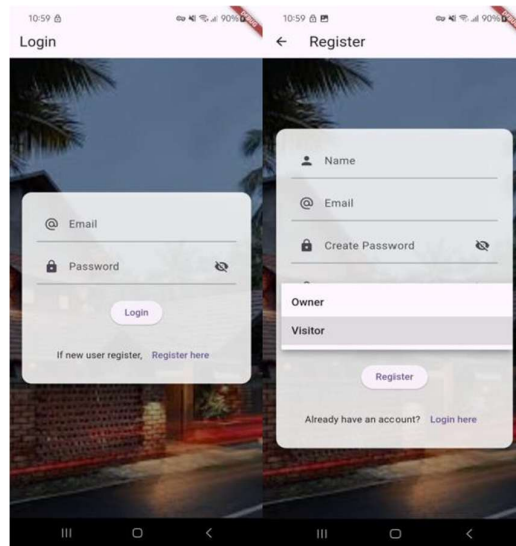
10. Home Rental App

UI/UX Wireframes or Screens

The UI design of the Home Rental App focuses on simplicity and ease of use. Below are the major screens (wireframes can be drawn in tools like Figma, Adobe XD, or Canva):

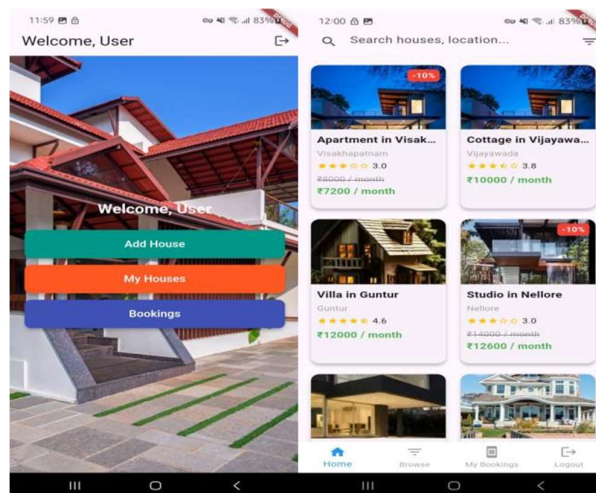
Login/Register Screen

- Fields: Email/Phone, Password.
- Buttons: Login, Register, Forgot Password.



Home Screen

- Top: Search bar.
- Sections: Categories
- Featured owners and offers.



house listing Screen

List of house items with images, prices, and “Add to wishlist” button.

house Item Details Screen

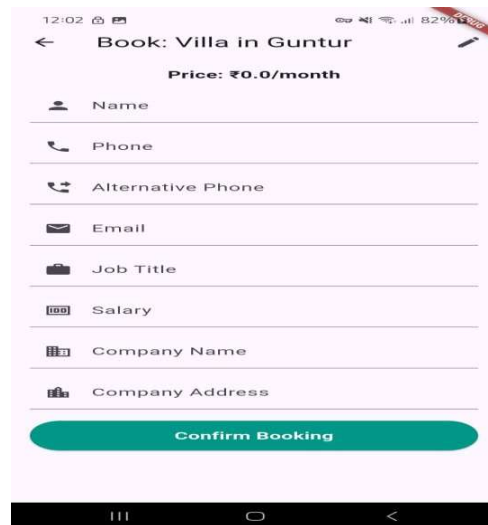
Large image, description, price, customization options



Wishlist Screen

List of selected items, quantity adjustment, total price.

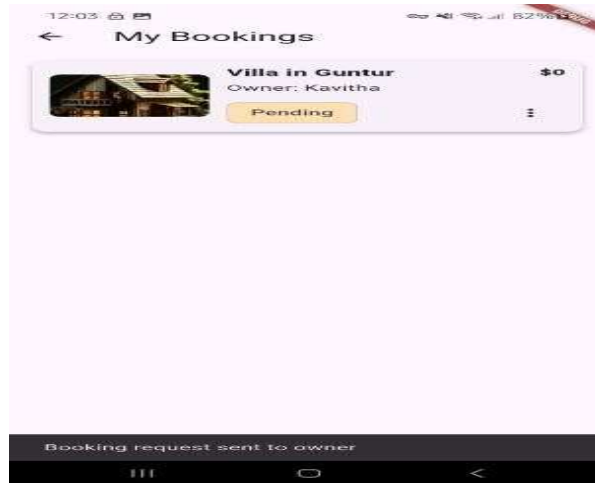
Checkout button.



Checkout Screen

confirmation address, payment option, confirm bookings button.

Bookings Confirmation Screen



Thank you message, estimated confirmation time, bookings ID.

bookings History Screen

Past bookings with status and rebookings opt

Implementation Details

The implementation of the Home Rental App focuses on building a cross-platform, scalable, and user-friendly system. It leverages modern mobile development frameworks, cloud services, and structured design patterns to ensure maintainability and smooth user experience.

Technologies Used (Extended)

- **Flutter (Frontend Framework)**
 - Flutter allows a single codebase to target both Android and iOS platforms, reducing development effort and cost.
 - It provides a reactive programming model, meaning the UI updates automatically when the underlying state changes.
 - The use of Material Design widgets (for Android look-and-feel) and Cupertino widgets (for iOS look-and-feel) ensures a native-like experience.
 - Hot Reload feature accelerates the development cycle by allowing real-time preview of code changes.
- **Dart (Programming Language)**
 - Dart's ahead-of-time (AOT) compilation ensures fast startup time for mobile apps.
 - Just-in-time (JIT) compilation makes debugging and development efficient.

- Provides null safety which prevents runtime crashes due to null values.
- Strong support for asynchronous programming makes it ideal for handling network requests and Firebase queries.

➤ **Firebase (Backend-as-a-Service)**

- Firebase Authentication → Enables multiple authentication methods (email/password, Google, phone number).
- Cloud Firestore → Provides a NoSQL document-based database with real-time synchronization, ensuring instant updates for bookings and wishlist changes.
- Firebase Storage → Stores images (house pictures, owners logos).
- Firebase Cloud Messaging (FCM) → Sends push notifications for bookings confirmation, preparation, and confirmation updates.
- Firebase Analytics → Tracks user behavior (popular house items, most bookings categories).

➤ **Third-Party APIs & Libraries**

- Payment Integration → Razorpay/Stripe APIs for secure transactions.
- Google Maps API → To display owners locations and track confirmation (optional).
- Provider/Bloc Packages → For state management.
- HTTP Package → For making REST API calls.

Key Components

➤ **Model (M)**

- Represents the data layer of the application.
- Includes APIs, databases, and business logic.
- Example in Flutter: Classes that fetch data from Firebase or REST APIs.

➤ **View (V)**

- The UI layer – what the user interacts with.
- Contains widgets, layouts, and animations.
- Example in Flutter: Scaffold, TextField, ListView.